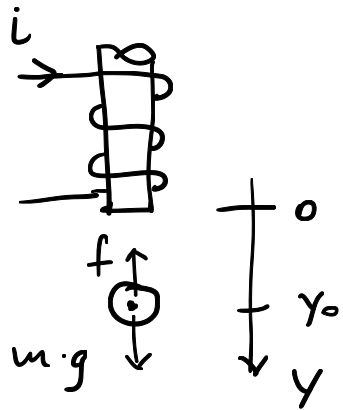


Linjärisering av ett system med flera variabler



$$f(i) = k \cdot \frac{i}{y^2}$$

$$-m \cdot \ddot{y} - f(i) + m \cdot g = 0$$

$$\begin{aligned} \Rightarrow m \cdot \ddot{y} &= -f(i) + m \cdot g \\ &= -\frac{k \cdot i}{y^2} + m \cdot g \quad (1) \end{aligned}$$

$$y = y_0 + \Delta y \Rightarrow \ddot{y} = \Delta \ddot{y}, \dot{y} = \Delta \dot{y}$$

$$i = i_0 + \Delta i$$

1. Arbetspunkten

$$\begin{aligned} \ddot{y} &= 0 \\ \Rightarrow \frac{k \cdot i_0}{y_0^2} &= m \cdot g \end{aligned}$$

$$\Rightarrow i_0 = \frac{m \cdot g}{k} \cdot y_0^2$$

2. Linjärisering

$$\frac{k \cdot i}{y^2} \approx \frac{k \cdot i_0}{y_0^2} + \left. \frac{\partial \left(\frac{k \cdot i}{y^2} \right)}{\partial y} \right|_{\substack{i=i_0 \\ y=y_0}} \cdot \Delta y +$$

$$\left. \frac{\partial \left(\frac{k \cdot i}{y^2} \right)}{\partial i} \right|_{\substack{i=i_0 \\ y=y_0}} \cdot \Delta i$$

$$= \frac{k \cdot i_0}{y_0^2} + \frac{-2k \cdot i_0}{y_0^3} \cdot \Delta y + \frac{k}{y_0^2} \cdot \Delta i$$

3. Skriv ut ekvationerna

$$(1) \Rightarrow m \cdot \Delta \ddot{y} = - \frac{k i_0}{y_0^2} + \frac{2 k i_0}{y_0^3} \cdot \Delta y - \frac{k}{y_0^2} \cdot \Delta i + m \cdot g$$

$$\Rightarrow m \cdot \Delta \ddot{y} = \frac{2 k i_0}{y_0^3} \Delta y - \frac{k}{y_0^2} \cdot \Delta i$$

4. Begynnelsevärden

$$\Delta i(t_0) = i(t_0) - i_0(t_0)$$

$$\Delta y(t_0) = y(t_0) - y_0(t_0)$$

